

Study Guide & Defense Q&A

Pneumonia-CVD GWAS-eQTL Colocalization | Bioinformatics 1 - Project 3 | crib sheet for the whole team

How to use: read Section 1 (concepts) and Section 2 (numbers to memorize) first - anyone may be asked these. Section 3 is the likely jury questions with model answers. Section 4 is a gene glossary.

1. Core concepts & definitions

SNP — Single-nucleotide polymorphism: a single-base position that varies between people. The unit of both GWAS and eQTL analysis.

GWAS — Genome-Wide Association Study: tests every SNP for association with a trait (here pneumonia, case vs control). Output per SNP: effect size (beta), standard error, p-value, allele frequency. Genome-wide significance = $p < 5e-8$.

Locus — A genomic region carrying an association signal, usually named after its lead gene (e.g. the CRP locus on chr1). A locus contains many correlated SNPs.

Lead SNP — The most strongly associated SNP at a locus (smallest p-value).

MAF — Minor Allele Frequency: frequency of the less common allele. Derived from allele frequency (if AF > 0.5, $MAF = 1 - AF$).

eQTL — Expression Quantitative Trait Locus: a SNP whose genotype is associated with the expression level of a gene, in a specific tissue. cis-eQTL = the SNP sits near the gene it regulates.

Colocalization — A statistical test asking whether a GWAS signal and an eQTL signal in the same region are driven by the SAME causal variant. If yes, the trait may act through that gene's expression.

coloc.abf / PP.H0-H4 — Bayesian colocalization (Giambartolomei 2014). Returns 5 posterior probabilities that sum to 1: H0 (no signal), H1 (GWAS only), H2 (eQTL only), H3 (both, DIFFERENT causal variants), H4 (both, SAME causal variant). We flag colocalization when $PP.H4 > 0.5$ (supervisor's threshold).

H3 vs H4 — The crux of our result. H4 = shared cause (the interesting case). H3 = the region has both a GWAS and an eQTL signal, but they are two different variants that happen to sit close together. H3 is a real, valid result - not a failure.

coloc.abf assumption — coloc.abf assumes ONE causal variant per region. At multi-signal loci this can force an H3 answer. coloc.susie relaxes this (handles multiple signals) but needs an LD reference panel.

LD (linkage disequilibrium) — Correlation between nearby SNPs (co-inherited). Dense LD is why signals cluster and why colocalization is needed rather than just eyeballing overlap. Extreme LD (MHC) makes coloc unreliable.

GTEx v8 / EBI eQTL Catalogue — GTEx = reference eQTL resource across human tissues. We used the EBI eQTL Catalogue (GTEx v8 reprocessed on GRCh38), queryable by region via API - instead of the 462 GB raw GTEx file.

FinnGen — Finnish biobank GWAS. We used Data Freeze 13, endpoint J10_PNEUMONIA (European ancestry, matches GTEx; genome build hg38).

Genome build (hg38) — Coordinate system for positions. GWAS and eQTL must share a build or SNPs mis-match. FinnGen DF13 and GTEx v8 are both hg38 -> no liftOver needed.

Harmonization — Making the two datasets comparable: match SNPs by position + allele pair, align the effect allele (flip eQTL beta if ref/alt swapped), drop ambiguous strand SNPs (A/T, C/G near MAF 0.5).

Case-control parameters — For a cc GWAS, coloc needs N (total sample), s (proportion of cases) and type='cc'. eQTL is type='quant'.

2. Numbers to memorize

GWAS

FinnGen DF13, J10_PNEUMONIA, hg38, 21.3M variants; 501 genome-wide significant ($p < 5e-8$)

Coloc params	N = 500,186 cases = 84,364 controls = 415,822 s = 0.1687 type = cc
eQTL tissues	blood QTD000356 (n=670), lung QTD000271 (n=510), liver QTD000266 (n=208) - all GTEx v8
Loci tested (4)	chr15 (140 hits), CRP/chr1 (82), HNF1A/chr12 (12), TNFSF15/chr9 (11)
Not testable	chr2 IFIH1 (1 hit, no eQTL), chr5 CDH6 (4 hits), chr6 MHC (251, excluded: extreme LD)
Master result	74 gene x tissue tests (blood/lung/liver), 0 colocalizations (max PP.H4 = 0.29, a lncRNA)
CVD overlap	list = 4,309 UNIQUE genes (file has 4,311 lines; 2 duplicates: CDKN2A, TNS1). ~20% of protein-coding. 7/16 GWAS-locus genes; 7/26 candidate genes
Validation	recovered HYKK + CHRNA3/5 at chr15 (matches published UKB+FinnGen pneumonia GWAS)

One-line result: Pneumonia and CVD share loci and genes, but colocalization finds no shared causal variants (PP.H3 approx 1 everywhere) - the overlap is at the gene/region level, not the causal-variant level, in blood, lung and liver.

3. Likely jury questions & model answers

Conceptual

Q What exactly does colocalization test, and why not just check if the same SNP is significant in both?

A. Colocalization tests whether a GWAS signal and an eQTL signal are driven by the same causal variant, accounting for LD. Two signals can share a top SNP by chance because nearby SNPs are correlated; coloc integrates over all SNPs in the region and returns the probability of a shared cause (PP.H4), which is far more rigorous than comparing single p-values.

Q What is the difference between H3 and H4? Why is your H3 result not a failure?

A. H4 means both traits are driven by the SAME causal variant (colocalization). H3 means both have a real signal but from DIFFERENT causal variants that sit close together. H3 is a genuine biological answer: it says pneumonia risk at that locus is not mediated by the cis-expression of that gene. It is informative, not a pipeline error - we recovered known genes, so the pipeline works.

Q What does PP.H4 = 0.29 (your best) actually mean?

A. It means only ~29% posterior probability of a shared causal variant at that gene (a lncRNA), well below the 0.5 threshold, with the remaining probability on H3. So even our strongest case favours distinct variants.

Methods

Q Why FinnGen rather than the other pneumonia GWAS (GCST90624053)?

A. FinnGen DF13 gives broader locus coverage (7 loci vs 3) and more immune/inflammatory genes relevant to the CVD link. The alternative is not independent - it already includes FinnGen R10 - so it is not a true replication. Prof. Yang validated keeping FinnGen (D1).

Q Why the EBI eQTL Catalogue instead of downloading GTEx directly?

A. The raw GTEx 'all associations' file is ~462 GB in a requester-pays bucket. The EBI Catalogue hosts the same GTEx v8 data, uniformly reprocessed on GRCh38 and queryable by region through an API, so we pull only the windows we need.

Q Why does genome build matter so much?

A. GWAS and eQTL positions must be on the same build or the same SNP has different coordinates and silently mis-matches, so coloc compares the wrong variants. FinnGen DF13 and GTEx v8 are both hg38, so no liftOver was needed - we verified this on a shared lead SNP.

Q Why remove ambiguous (A/T, C/G) SNPs near MAF 0.5?

A. For these SNPs you cannot tell which strand an allele is on, and near MAF 0.5 you cannot use frequency to resolve it, so the effect-allele alignment between datasets becomes unreliable. We drop them to avoid sign errors in beta.

Q Why exclude the MHC (chr6)?

A. The MHC has extreme, long-range LD; colocalization assumptions break down there and results are unreliable. It was our largest signal (251 hits) but we excluded it by design and say so explicitly.

Q Why blood and lung, and then liver?

A. Whole blood is the shared immune readout for all traits; lung is the site of pneumonia. We added liver on the supervisor's suggestion because CRP and HNF1A are primarily liver-expressed and lacked blood/lung eQTLs.

Results & interpretation

Q Does your negative result disprove the shared-genetics hypothesis?

A. No. We show that at the tested loci, in the tested tissues, the shared architecture is at the gene/region level, not the same causal variant acting via cis-expression. The hypothesis could still hold through other tissues, trans-effects, multiple signals (susie), or non-expression mechanisms. We are careful to state this.

Q Why do CRP and HNF1A - your best hypothesis genes - have no eQTL even in liver?

A. GTEx does not report a significant common cis-eQTL for these genes in blood, lung or liver. Some genes (especially tightly or inducibly regulated ones like CRP) simply lack common cis-eQTLs, and GTEx liver has a smaller sample (n=208), reducing power. So their lead genes remain untestable - which we present as a demonstrated limitation across three tissues.

Q Isn't the CVD list 4,311 genes? (the plan says so)

A. The file has 4,311 lines, but two symbols are duplicated - CDKN2A and TNS1 - so there are 4,309 UNIQUE genes. We deduplicated, so every figure uses 4,309 (the Venn shows 4,302 CVD-only + 7 shared).

Q The CVD list has 4,309 genes - isn't 7/16 overlap just chance?

A. Exactly the caveat we state. With ~20% of all protein-coding genes on the list, some overlap is expected by chance, so an overlap is suggestive, not proof. It is strengthened only by (1) a high PP.H4 - which we do not have - or (2) a clear biological story (CRP, HNF1A, P2RX4). We deliberately do not overclaim.

Q How do you know your pipeline is correct?

A. We recovered the known 15q25 pneumonia genes HYKK and CHRNA3/5, which match published UK Biobank + FinnGen results. Recovering established biology is our positive control.

Challenging / statistical

Q Could the lack of colocalization just be low power?

A. Partly possible, and we say so. But the tested loci had strong GWAS signals (down to $p \sim 2e-24$) and strong eQTLs, and coloc still returned H3 approx 1 - a confident 'distinct variants' answer, not an 'uncertain' one. Power is a bigger concern for the untestable loci (IFIH1, CDH6), which we did not over-interpret.

Q coloc.abf assumes one causal variant - could that hide a real colocalization?

A. Yes, that is the main methodological limitation. At multi-signal loci (chr15, CRP, HNF1A) H3 can mask an H4. The rigorous follow-up is coloc.susie, which handles multiple signals but needs an LD reference panel (e.g. 1000 Genomes). The supervisor agreed this is future work, not required here.

Q Did you correct for multiple testing across 74 tests?

A. Colocalization is not a null-hypothesis test with p-values to correct; it reports a posterior probability per gene. We apply the fixed PP.H4 > 0.5 threshold across all tests, and none pass. We could also report a stricter 0.8 threshold - it changes nothing here.

Q You used nearest-gene annotation - could you have missed the real causal gene?

A. Possibly. Nearest-gene is a first-pass label; the true target can be further away. That is precisely why we run eQTL colocalization per gene in the window rather than trusting the nearest gene - and why fine-mapping / susie would be the next refinement.

4. Gene glossary (say one sentence if asked)

CRP	C-reactive protein; classic acute-phase inflammation marker and a CVD risk biomarker. Lead gene of the chr1 locus; no blood/lung/liver eQTL.
HNF1A	Transcription factor; regulates CRP and is an established coronary-artery-disease gene; mainly liver-expressed. chr12 lead; no eQTL in our tissues.
TNFSF15	TNF superfamily member (TL1A); inflammation and endothelial/vascular biology. chr9 lead; has a strong blood eQTL but still H3.
P2RX4	Purinergic receptor; inflammation (NLRP3 inflammasome) and endothelial function - a plausible inflammation-to-CVD bridge. chr12 candidate (H3).
CTSH	Cathepsin H; antigen processing and immunity. chr15 candidate present in blood and lung.
AIM2	Innate-immune DNA sensor (inflammasome). chr1 candidate.
ORM1	Orosomucoid / alpha-1-acid glycoprotein; acute-phase inflammation protein. chr9 candidate.
SLAMF8 / IGSF8	Immune signalling receptors in the chr1 region.
CHRNA3 / CHRNA5 / HYKK	15q25 smoking-related locus; our validation that the pipeline recovers known pneumonia genes.
IREB2	Iron-regulatory gene at 15q25.
IFIH1	MDA5, viral-RNA sensor (innate immunity). chr2 - biologically ideal but had only 1 GWAS hit and no cis-eQTL, so untestable.

Golden rule for the defense: if unsure, restate the honest headline - "we find shared genes/regions but not shared causal variants; H3 is a real result, and coloc.susie plus more tissues are the next steps."